GUHNOO YUN

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RESEARCH STATEMENT

Analyzing and designing the inner workings of AI perception models to improve how machines see and understand the real world. AI perception models can work well long before anyone knows why. My research explores the underlying mechanisms, often by treating each core component inside a model as a filter and analyzing what it keeps or removes, from broad patterns to fine details, as information passes through it. I then turn that understanding into better designs.

- Analyzed convolution and attention as spectral filters, and designed new components that balance fine details and broad patterns. Then grounded them in theory, and the resulting models outperformed leading vision architectures. [1, 4]
- Viewed linear attention as a fixed-size memory, whose cost grows only in proportion to input size, and used a filter to decide what that memory keeps and recalls. The result is a general-purpose operation that works across image classification, object detection, segmentation, and image generation. [11]
- Developed vision algorithms that detect falls and recognize daily activities for care robots at home and in hospital isolation wards. [5, 6, 8, 9] Built the vision part of systems pairing first-person video with brain signals, using foundation models and large language model agents to describe where a person is and what they came to do. [2, 3]

Looking ahead, I aim to unify the different mathematical views of AI perception models, such as filters and memories, into a single framework that brings machines closer to human intelligence. My work already builds on both of these views. With this framework, I will design models that combine video, sound, and body signals, and that run well on the chips inside real machines such as humanoid robots. The goal is machines that understand the real world, built on designs we can explain.

EDUCATION

Feb 2027 (expected) **Ph.D. in Computer Science and Engineering**
Korea University Seoul, Korea
Dissertation: *Spectral Analysis and Adaptive Filtering for Visual Perception*

2018 **M.S. in Mechatronics**
Gwangju Institute of Science and Technology (GIST) Gwangju, Korea
Thesis: *Removal of Jitter Noise using a Modified Kalman Filter in 3D Shape Recovery from Focus*

2016 **B.S. in Control and Measurement Engineering**
Gyeongsang National University (GNU) Jinju, Korea

RESEARCH EXPERIENCE

Since 2019 **Student Researcher**
Intelligence and Interaction Research Center, **Korea Institute of Science and Technology (KIST)**
Supervisor: Dr. Dong Hwan Kim
Developed vision architectures and efficient attention [1, 4, 11], vision algorithms for care robots and first-person video [2, 3, 5, 6], and 3D and multi-view vision methods [7, 10].

2018 – 2019 **Intern Researcher**
Center for Intelligent Robotics, **KIST**
Developed vision algorithms for the care of older adults at home, such as detecting falls and monitoring people with dementia. [8, 9]

2016 – 2017 **M.S. Student**
Department of Mechatronics, **Gwangju Institute of Science and Technology (GIST)**
Advisor: Prof. Tae-Sun Choi
Worked on 3D shape recovery from images taken at different focus settings.

Aug 2015 **Summer Undergraduate Research Fellowship**
Computer Science and Engineering, **Ulsan National Institute of Science and Technology (UNIST)**
Supervisor: Prof. Jaesik Choi
Studied deep learning.

PUBLICATIONS

Published / Accepted

[1] Guhnoo Yun, J. Yoo, K. Kim, J. Lee, P.H. Seo, D.H. Kim, **Spectral-Adaptive Modulation Networks for Visual Perception.** *IEEE Trans. Pattern Anal. Mach. Intell. (TPAMI)* 48(10), 11511–11525 (2026). [doi: 10.1109/TPAMI.2026.3690455] [arXiv] [code]

[2] W. Hyung, M. Kim, Y.-S. Kim, J. Lee, B. Ko, Guhnoo Yun, D.H. Kim, C.-H. Im, **Context-Aware Daily-Life Arousal Monitoring Using Egocentric Vision and EEG.** *Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)* (2026). [program]

[3] Y.-S. Kim, M. Kim, W. Hyung, B. Ko, Guhnoo Yun, D.H. Kim, C.-H. Im, **PeBCI: Vision-Guided Brain-Computer Interface for Context-Aware Neural Decoding in Daily Life.** *Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)* (2026). [program]

[4] Guhnoo Yun, J. Yoo, K. Kim, J. Lee, D.H. Kim, **SPANet: Frequency-balancing Token Mixer using Spectral Pooling Aggregation Modulation.** *IEEE/CVF Int. Conf. on Computer Vision (ICCV)*, 6090–6101 (2023). [doi: 10.1109/ICCV51070.2023.00562] [CVF] [arXiv] [project] [code]

[5] Y. Kwon, S. Shin, K. Yang, S. Park, S. Shin, H. Jeon, K. Kim, Guhnoo Yun, …, Y. Lim, **Heterogeneous Robot-Assisted Services in Isolation Wards: A System Development and Usability Study.** *IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS)*, 8069–8076 (2023). [doi: 10.1109/IROS55552.2023.10341857]

[6] K. Kim, Guhnoo Yun, S.-K. Park, D.H. Kim, **Efficient Fall Detection for a Healthcare Robot System Based on 3-Axis Accelerometer and Depth Sensor Fusion with LSTM Networks.** *Int. Conf. on Pattern Recognition (ICPR)*, 2207–2212 (2022). [doi: 10.1109/ICPR56361.2022.9956418]

[7] H.-S. Jang, Guhnoo Yun, M.T. Mahmood, M.-K. Kang, **Optimal Sampling for Shape from Focus by Using Gaussian Process Regression.** *IEEE Int. Conf. on Consumer Electronics (ICCE)*, 1–4 (2020). [doi: 10.1109/ICCE46568.2020.9043150]

[8] K. Kim, Guhnoo Yun, S.-K. Park, D.H. Kim, **Fall Detection for the Elderly Based on 3-Axis Accelerometer and Depth Sensor Fusion with Random Forest Classifier.** *Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, 4611–4614 (2019). [doi: 10.1109/EMBC.2019.8856698]

[9] Guhnoo Yun, K. Kim, S.-K. Park, D.H. Kim, **A Monitoring System to Support Home Health Care for the Elderly with Dementia by Detecting Going Out Activities Based on RGB-D Sensors.** *Int. Conf. on Ubiquitous Robots (UR)*, 71–76 (2019). [doi: 10.1109/URAI.2019.8768598]

Under Review

[10] Y. Lee, B. Ko, Guhnoo Yun, D.H. Kim, **Every View Counts: View-Consistent Panoptic Quality for Multi-view Panoptic Segmentation.** *arXiv preprint* (2026). [arXiv: 2610.05911]

In Preparation

[11] Guhnoo Yun et al., **SALAFormer: Spectral-Adaptive Linear Attention as Closed-Form Associative Memory.**

PATENTS

2025 **Frequency-Balancing Token Mixer and Its Backbone Model**
Korean Patent No. 10-2815184 (registered May 27, 2025)

2022 **System and Method for Determining Whether a Resident Has Left Home**
Korean Patent No. 10-2374057 (registered Mar 8, 2022)

AWARDS & SCHOLARSHIPS

2026 **Outstanding Student Researcher Scholarship**
Korea Institute of Science and Technology (KIST)

2023 **Outstanding Technology Award**
Korea Institute of Science and Technology (KIST)

2020 **Presidential Award** 2020 International Robot Contest—Humanoid Robot Sports (Deep Learning Algorithm)
Ministry of the Interior and Safety

2016 – 2017 **Graduate Scholarship**
Gwangju Institute of Science and Technology (GIST)

2015 **Grand Prize** 2015 Capstone Design Competition
Gyeongsang National University (GNU)

2013 – 2015 **Academic Excellence Scholarship**
Gyeongsang National University (GNU)

INVITED TALKS

2026 **2026 Inter-Ministerial Public Technology Transfer and Commercialization Roadshow**
Presentation at the Science and Technology Convention Center, Seoul

2021 **"Recent Research Trends in Computer Vision"**
Seminar at Korea Polytechnic University (now Tech University of Korea)

2021 **"6D Object Pose Estimation"**
Seminar series at CUBOX (now SECERN AI)

PROFESSIONAL SERVICE

2024, 2026 **Reviewer, International Conference on Pattern Recognition (ICPR)**

TECHNICAL SKILLS

Languages & Frameworks: Python, PyTorch, C++, MATLAB; timm, MMDetection, MMSegmentation, ROS

REFERENCES

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Dr. Yoonseob Lim
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